



C23-EC-105

23057

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FIRST YEAR EXAMINATION

ELECTRONIC COMPONENTS AND DEVICES

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State the factors affecting the capacitance of a capacitor.
2. List any three applications of RF chokes.
3. State the necessity of connectors in electronic circuits.
4. Distinguish between intrinsic and extrinsic semiconductors.
5. List any three applications of varactor diode.
6. Draw the circuit symbols of NPN and PNP transistors.
7. List any three advantages of JEET over BJT.
8. State the working principle of photo diode.
9. Write any three applications of opto-coupler.
10. List the specifications of microphones.

PART—B

10×5=50

- Instructions :**
- (1) Answer *any* **five** questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** With a neat sketch describe the working of Rheostat and give its applications.
- 12.** Explain the working of general -purpose electromagnetic relay and mention applications of relays.
- 13.** Explain energy band diagrams of conductors, semiconductors and insulators.
- 14.** Explain the working of PN junction diode with forward and reverse biasing.
- 15.** Derive the relations among alpha, beta and gamma factors of a transistor.
- 16.** Draw the explain the drain characteristics of JFET.
- 17.** Explain the structure, working principle and characteristics of LED.
- 18.** Explain the construction and working of Horn loud speaker with suitable diagram.

★ ★ ★